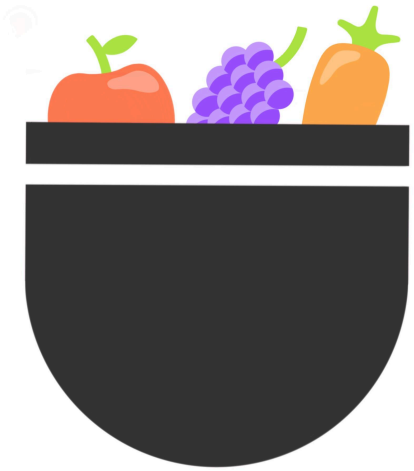

Pocket Pantry

Project Proposal

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Table of Contents

1 Executive Summary **3**

2 Introduction **4**

2.1 Problem Background 4

2.2 Needs Statement 4

2.3 Goals and Objectives 5

3 Research **6**

3.1 User Stories 6

3.2 Current Solutions 6

4 Proposed Work **11**

4.1 Algorithms 11

4.2 Data 14

4.3 UI Layout 15

4.4 Scope 16

4.5 Tech Stack 17

5 Engineering Standards **17**

5.1 Project Management 17

5.2 Team Roles 18

5.3 Schedule of Tasks 19

6 References **20**

1 Executive Summary

The Pocket Pantry project addresses food management issues faced by college students. They often struggle with tight budgets and busy schedules which leads to unsustainable food management. Using an Android mobile application, Pocket Pantry will help these students efficiently manage their food inventory, minimize waste, and save money.

College students buy groceries without fully utilizing existing ingredients, leading to food waste and unnecessary expenses. Additionally, conflicts often arise in shared living situations over personal versus communal food items. The meal planning applications that currently exist do not adequately address these specific needs.

Pocket Pantry will offer four core functionalities:

1. Inventory Management: Users can track ingredient quantities and expiration dates, ensuring they prioritize the use of perishable items.
2. Shared Ingredient Organization: The app will allow users to label items as personal or shared, reducing conflicts and duplicate purchases among roommates.
3. Meal Recommendations: Based on available ingredients and dietary preferences, the app will suggest recipes that minimize waste and optimize meal planning.
4. Cost-Saving Features: Users will receive recommendations for budget-friendly recipes and grocery items, helping to alleviate financial pressures.

The main goal of Pocket Pantry is to develop an intuitive mobile application that effectively manages food inventory and enhances meal planning. Objectives include implementing an expiration date ranking algorithm, creating a SQL database for user inventory and recipe data, receipt scanning, and automated shopping lists.

While there are various meal planning applications, most of them either lack comprehensive features or charge users for core functions. Pocket Pantry distinguishes itself by integrating pantry management with recipe suggestions while remaining free to users.

The project will be managed using GitHub projects for task tracking, with tasks being divided between developers (front and back end), a project manager, and a graphics designer. The initial focus will be on Android development, with potential plans for cross-platform support in the future.

Pocket Pantry aims to help college students by providing them with the tools to effectively manage their food resources, reduce waste, and make informed financial decisions. This will ultimately help them in leading a more sustainable lifestyle.

2 Introduction

2.1 Problem Background

College students often face tight budgets and busy schedules, making it difficult to efficiently manage their food expenses and plan meals. As a result, many students buy groceries without fully utilizing what they already have, leading to ingredients expiring before they are used. Additionally, those living with roommates may have trouble differentiating between shared and personal ingredients, causing duplicate purchases and confusion over what belongs to each person. This food waste adds an extra financial strain on students who may already be struggling to make ends meet. Our proposed solution will address these challenges, helping students reduce costs by minimizing food waste.

2.2 Needs Statement

While there are a variety of meal planning applications available, most do not specifically cater to the unique needs and constraints of college students. With that in mind, our proposed solution will focus on four key need areas: efficient inventory management with expiration tracking, organizing shared and personal ingredients, providing simple, time-saving meal recommendations, and minimizing costs.

Effective inventory management is essential for college students to ensure that they make the most of their existing ingredients and new purchases. To prevent food waste, students need a reliable method to track expiration dates, helping them prioritize the use of perishable items before they go bad. Additionally, using existing ingredients in recipes may lead to more leftover ingredients, perpetuating the problem.

College students living with roommates need a way to differentiate between shared and personal ingredients. Currently, there are no resources out there that directly address this problem. Organizing these items can help minimize ownership conflicts and duplicate purchases.

Streamlined meal planning tools are necessary for busy college students, allowing them to quickly create meals that utilize their existing ingredients. Additionally, students need a way to schedule their meals around their personal commitments and activities.

Given their financial challenges, college students require ways to cut expenses when meal planning. Students need a way to identify cost-effective recipes that utilize their existing ingredients while also receiving suggestions for budget-friendly grocery items.

2.3 Goals and Objectives

Our proposed solution, *Pocket Pantry*, will be a mobile application. Our primary goal is to address the aforementioned key areas: managing existing stock, organizing shared inventory spaces, planning meals, and minimizing costs.

Effective inventory management is crucial for optimizing ingredient usage and minimizing food waste. To facilitate this, we will track expiration dates, allowing users to monitor ingredient freshness and prioritize the use of items that are nearing their expiration. Users can also manually input items into their inventory, creating a personalized experience that accurately reflects what they have on hand.

Our application will support shared inventory spaces to help roommates differentiate between shared and personal ingredients. A table in the cloud will be keeping track of shared ingredients between multiple users, and each users' individual inventory will be automatically updated based on what is marked as shared.

Our meal planning tool is designed to save time and minimize costs while promoting efficient ingredient use. Recipes will be recommended based on an expiration date ranking algorithm, encouraging users to utilize ingredients that are close to expiring. Users can also add meals to their calendar, allowing them to organize their meal schedule around personal commitments, and can save recipes to a personal library for easy access and reuse. Additionally, the application will provide suggestions for budget-friendly meals and grocery items.

Our mobile application will be specifically developed for Android only due to time constraints; however, we plan on leaving the door open for future cross-platform support. We opted to target Android over IOS because it has no publisher fee and offers better support for development tools.

We plan to implement a set of additional minor features that augment the core functionality of Pocket Pantry. These features include receipt scanning, shopping list creation, automatic inventory updates based on cooked meals, and recipe filters (allergies, dietary preferences, ingredient exclusions/inclusions, macronutrient levels, etc.).

3 Research

3.1 User Stories

As a student juggling studies and work, I want to create a meal plan based on the available ingredients and my schedule, so I can optimize my time and food usage.

As a college student sharing a kitchen with roommates, I want to easily mark items as personal or shared, so we can better manage grocery purchases and avoid conflicts.

As a busy college student, I want to receive reminders when ingredients in my pantry are about to expire, so I can prioritize using them before they go bad.

3.2 Current Solutions

A variety of apps have already been created to address the issues of recipe recommendations, pantry management, and food waste. These solutions attempt to simplify the process of meal planning and pantry tracking. While many of the platforms offer useful features, they often fall short in certain areas. Meal planning focused apps lack features for pantry management, while pantry tracking apps fail to use their internal data representation to effectively recommend recipes. For the atypical app that offers both, they lock this functionality behind a paywall. Despite working solutions existing, there is significant room for improvement in creating an integrated experience between meal planning and pantry management. Below we will take a more in depth look at a handful of solutions that currently exist.

Mealime

Mealime is a meal planning app that helps users create personalized meal plans based on their dietary preferences, nutrition goals, and “ingredient availability”. The app generates shopping lists from the selected recipes and allows the user to order grocery delivery directly in-app. They market themselves as “meal planning made easy”. They “make meal planning, grocery shopping, and cooking as simple as getting takeout”.

While Mealime does make meal planning and grocery ordering easy, their claim of using ingredient availability is misleading. They allow you to search recipes that contain individual ingredients but don’t take into account the entirety of what's in your pantry to suggest meals. This results in the end user having to manually cycle through the ingredients they want to use and mentally keeping track of which meals use which ingredients when deciding how to use their ingredients on hand.

Pocket Pantry will utilize our internal representation of a user's fridge and pantry to recommend meals that prioritize the use of ingredients nearing their expiration date, taking into account uncommon quantities.

What Mealime does well

- Meal plans based on dietary preferences and individual ingredients
- "Wide variety" of "easy to make" recipes, according to Mealime
- Grocery list can be ordered in-app; however, can lead to over ordering and **more** waste

What Pocket Pantry does better

- Meal plans based on dietary preferences, a user's entire fridge & pantry, and ingredient quantities
- Upwards of 180,000 recipes with 700,000 real user reviews, including varying levels difficulties
- Cross references grocery list with inventory to track usage automatically
- Ensures only missing/needed ingredients are added to shopping list
- Helps users **avoid** over ordering

Yummly

Yummly is a popular app for personalized recipe recommendations and meal planning. It is a direct competitor with Mealime. As a result, they share many similar features. Yummly's core functionality is in their meal recommendation engine coupled with highly detailed instructions including photos. They call themselves a "smart cooking sidekick".

Yummly serves a user base that wants highly detailed instructions and a capable meal recommendation engine based on their previous meals. Yummly doesn't track a user's pantry and utilize that information which leaves that up to the user.

Pocket Pantry isn't as focused on the detailed instructions with images. We offer step by step instructions for every meal; however, don't provide images due to the overhead of storing tons of pictures. Additionally- reusing generic pictures for ingredients doesn't offer the user much real value in the end for their meal.

What Yummly does well

- Recipe recommendations based on patent pending "Food Intelligence"

What Pocket Pantry does better

- Recipe recommendations using ingredients from the user's pantry to minimize food waste

- “Extensive” recipe database
- Provides a shopping list based on chosen recipes.

- Over 180,000 recipes with varying levels of complexity
- Provides a shopping list based on chosen recipes and quantity of items in storage
- Automatically removes items from inventory when meals are made

Pantry Check

Pantry check is the top app for pantry tracking. They offer ingredient management with expiration date alerts to keep you on top of your expiring food. They offer restock suggestions with accurate prices, estimation of best-by dates, and inventory management “straight from the lock screen”. In addition, they also offer receipt scanning to make updating your inventory easier.

While Pantry Check offers some nice features like receipt scanning, they put limits on how many items you can scan before having to upgrade to the paid version, essentially transforming the app into a glorified spreadsheet. Pocket Pantry’s receipt scanning won’t have hard limits.

In addition to the inventory management features, pocket pantry utilizes the built in inventory representation required for a management app to recommend meals based on a variety of factors, while pantry check is solely a pantry tracker.

What Pantry Check does well

- Receipt scanning with a hard limit
- Restock suggestions with pricing
- Per item expiration dates

What Pocket Pantry does better

- Unlimited receipt scanning
- Shopping list with concrete quantities
- Per item expiration dates with the ability to freeze or manually modify

Kitchen Pal

Kitchen Pal is an all in one kitchen management app that helps users manage their pantry, reduce food waste, and plan meals. Their key features include a pantry management system, a grocery list, recipe suggestions based on ingredients in stock, along with nutritional information. While their features sound extensive, a considerable portion of the core functionality is locked behind a paywall. Their meal planning, barcode scanning, and custom kitchen storage sections all require a subscription to the app.

Pocket Pantry offers functionality like meal planning based on ingredients and custom inventory organization free of charge. For users who want the robust features of a food management tool along with the capability to plan meals and save money by using ingredients in stock, Pocket Pantry is the obvious choice- providing these tools at no cost to the user, making it the better choice for the budget-conscious consumer.

What Pantry Pal does well

- Recipe suggestions based on user's ingredients if subscription purchased
- Unlimited barcode scanning if a subscription is purchased
- Custom kitchen sections if subscription purchased

What Pocket Pantry does better

- Recipe suggestions based on user's ingredients
- Unlimited receipt scanning
- Custom kitchen sections

Feature Comparison

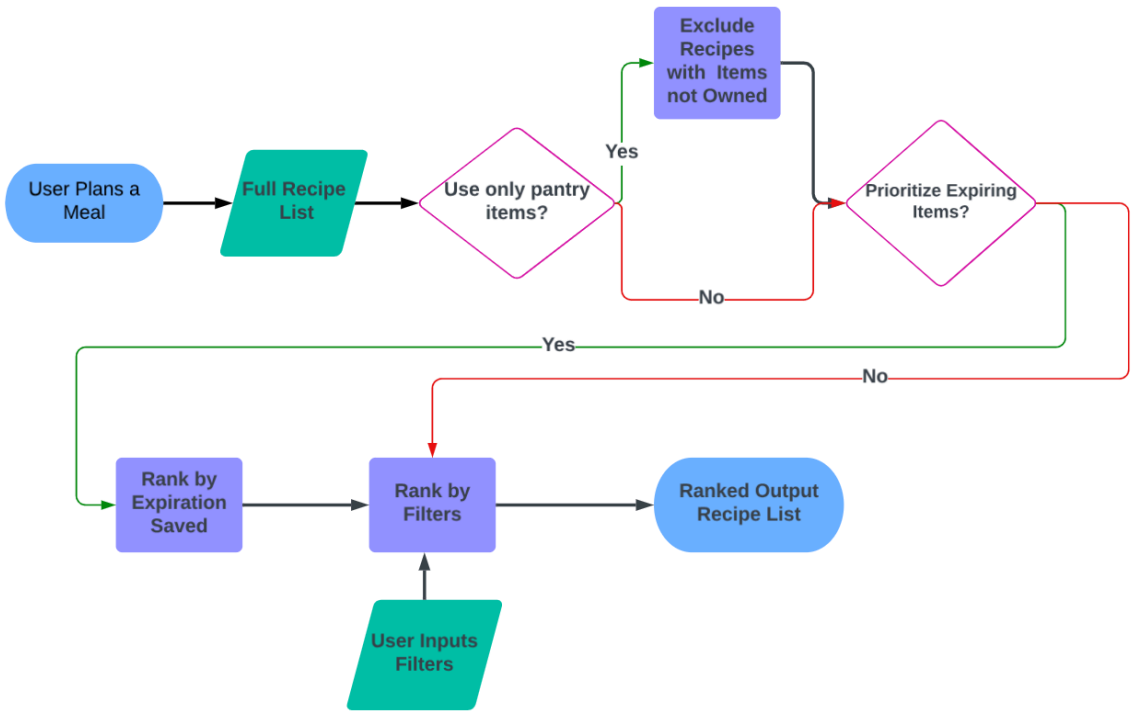
Feature / App	Pocket Pantry	Mealime	Yummly	Kitchen Check	Kitchen Pal
Pantry Management	✓	✗	✗	✓	✓
Recipe Suggestions	✓	✓	✓	✗	✗
Recipe Suggestions w/ current inventory	✓	✗	✗	✗	$
Receipt / Barcode scanning	✓	✗	✗	$	$
Expiration Date Tracking	✓	✗	✗	✓	✓
Automated Shopping list	✓	✓	✓	✗	✓
Shared inventory	✓	✗	✗	✗	✗

4 Proposed Work

4.1 Algorithms

Our app will utilize 3 main algorithms for its core functionality:

- `shopping_list_generator()` creates a shopping list based on planned meals
- `filter_by_user_preferences()` filters recipes based on user preferences (diet, allergies, macro/micronutrient quantities, etc.)
- `sort_recipes_expr_date()` sorts recipes based on their efficacy of "saving" ingredients
- `recipes_from_inventory()` filters recipes to ones that can be made with only ingredients user has on hand



Algorithm Flow Chart

The flow chart above provides a high level overview of the Pocket Pantry user interaction for generating desired recipes. On the recipe page of our app users will be able to filter recipe options based on current home inventory or not. The first algorithm in the flow chart will then pare down the full list of recipes to those that the user can make with only items on hand. The user will also make the decision to prioritize the listing of recipes based on the ratio of minimum

expiration time to maximum items saved. If yes is selected the listing of recipes will be ranked based on the algorithm `sort_recipes_expr_date()` to help the user select from recipes that will assist with managing their food waste. Finally the recipes will be pared down once again by our algorithm below based on their selected preferences (diet, allergies, macro/micronutrient quantities, etc.). The list of recipes will then be outputted to the user for them to choose from based on their inputs.

Shopping list generation algorithm pseudocode

```
function shopping_list_generator(planned_recipes, inventory)
   planned_quantities = {}
   for every recipe in planned_recipes
       for ingredient in recipe
           planned_quantities[ingredient.name] += ingredient.quantity

   shopping_list = {}
   for pair(ingredient_name, planned_quantity) in planned_quantities
       difference = inventory.quantity(ingredient_name) - planned_quantity
       if difference < 0
           shopping_list[ingredient_name] += -difference

   return shopping_list
```

User preference filtering algorithm pseudocode

```
function filter_by_user_preferences(recipes, user_preferences)
   for recipe in recipes
       for preference in user_preferences
           if recipe does not meet preference
               remove recipe from recipes
```

Sorting recipes by expiration date algorithm pseudocode

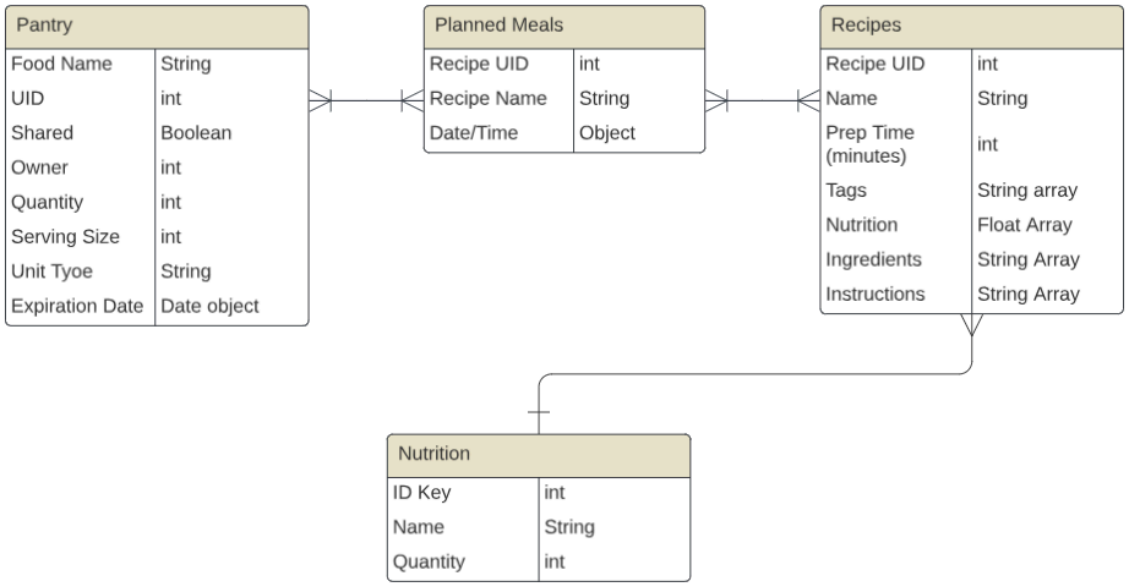
- Top recipes are ones that minimize the total expiration time while maximizing number of ingredients used
- Expiration time is days until food is bad, computed based on expiration date
- if you have 1 onion with 1 day, or (1 chicken with .25 days and 1 lemon with .5 days) -> prioritize the chicken and lemon because chicken.expiration_time + lemon.expiration_time < onion.expiration_time and it uses 2 ingredients as opposed to 1

```
function sort_recipes_expr_date(recipes, inventory)
   sorted_inventory = sort_by_expiration_time(inventory)
   recipe_scores = {}
   for recipe in recipes
       total_expiration_time = 0
       matched_ingredients = 0
       for ingredient in recipe
           pantry_item = find_soonest_expiring(sorted_inventory,
ingredient)
           if pantry_item != null
               total_expiration_time += pantry_item.expiration_time
               matched_ingredients += 1
       # lower expiration time is better -> soon to expire items
 should be prioritized
       # higher matched ingredients is better -> more ingredients
 used from pantry
       recipe_score = (matched_ingredients / total_expiration_time)
       recipe_scores[recipe.name] = recipe_score
   sorted_recipes = sort_by_score(recipe_scores)
   return sorted_recipes
```

Recipes from only inventory items algorithm pseudocode

```
function recipes_from_inventory(recipes, inventory)
   valid_recipes = []
   for recipe in recipes
       can_make_recipe = True
       for ingredient in recipe
           if ingredient not in inventory
               can_make_recipe = False
               break
       if can_make_recipe
           valid_recipes.append(recipe)
   return valid_recipes
```

4.2 Data



SQL Database UML Diagram

The diagram above provides a rough structure of the attributes and item types for the tables we plan on using for our PostgreSQL and SQLite3 backend data storage. It consists of 4 tables: Pantry, Planned Meals, Recipes, and Nutrition.

Recipes

Our recipe database is going to be built from a kaggle dataset containing food.com data. This dataset contains over 180,000 recipes along with more than 700,000 user reviews. In addition to a recipe's required ingredients, the dataset also contains the amount of time a recipe takes, nutritional information, user submitted tags, along with the required steps for cooking. The database we build on top of this dataset will be the core recipe bank that is used in filtering and processing when a user plans a meal.

Pantry

Our pantry will be a table that keeps track of food names, quantities, units, expiration dates among other fields. This pantry table will be used heavily by the algorithms mentioned in section 4.1. This pantry table will be read from to display relevant quantities in our pantry view.

Planned Meals

Planned meals is our table where we will store meals that the user plans. This table will be used with the pantry table when building out the users shopping list in its respective view. When a

planned meal is cooked (manually checked off by the user or automatically removed as the date passes, based on user preference) ingredients' respective quantities will be automatically subtracted from the users inventory.

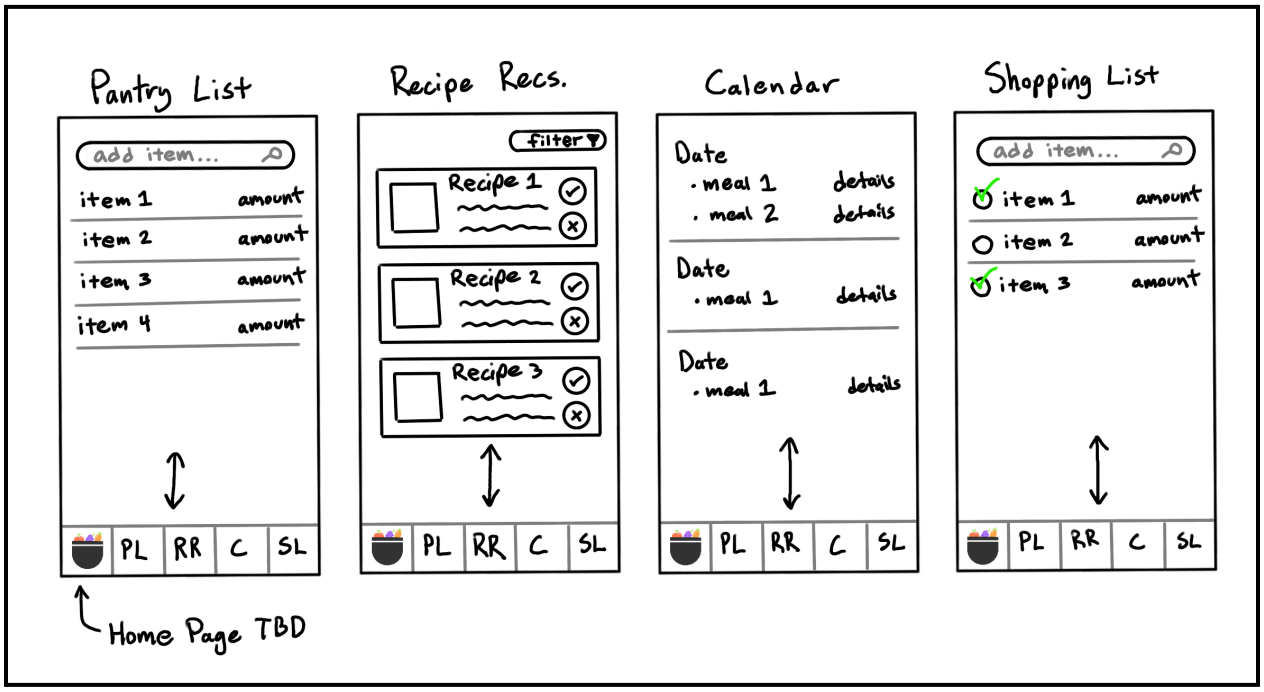
Nutrition

Nutrition is a small table which will store the recommended daily value of macro nutrients. The nutrition info provided in our dataset is given in percent daily value, which doesn't mean much in terms of actual quantities. We are still deciding between converting the nutrition information in our database to quantities or doing it on the fly with a small table. Our current database layout shows this nutrition information table.

4.3 UI Layout

Our App will have 5 screens:

1. Home Page (not shown in figure)
2. Pantry List
3. Recipe Recommendations
4. Calendar View
5. Shopping List



The Pantry List screen will be used for managing what students have in their pantry and fridge. The screen will list stored ingredients, quantities, expiration dates, and note if an item is shared

among all household members. Students will also be able to manually add and remove items from their inventory.

The Recipe Recommendations screen will recommend recipes to students based on ingredients they have in their inventory. Users will be able to customize recipe recommendations using filters. Currently, we plan to only allow filtering based on current inventory, expiration dates, and allergies. However, we hope to add more customizability to recipe recommendations if time allows, such as by calories or cost. Students can select recipes from this screen to plan future meals and automatically update their shopping list.

The Calendar View screen will allow users to see, manage, and cancel their planned meals. Users can also mark a meal as cooked, which will automatically update their inventory.

The Shopping List screen will auto populate based on selected recipes. Users can also manually add items to their shopping list. When an item is checked off from the shopping list, it will automatically be added to a user's inventory.

4.4 Scope

What Pocket Pantry is	What Pocket Pantry isn't
● A mobile app designed to help college students manage their food inventory and save money by preventing food from expiring	● A general-purpose meal planning app
● Focused on organizing individual and shared ingredients	● A shopping app or marketplace for purchasing groceries
● Provides meal recommendations based on ingredients users already have	● A complex gourmet recipe app or a platform offering advanced culinary skills
● Allows users to filter down recipes based on allergies, nutrition, etc.	● Dependent on AI-driven features
● Incorporates receipt scanning for easy addition of inventory ingredients	● A fully automated app that pulls grocery data without user interaction

4.5 Tech Stack

React native

Our front end will be built using react native. Our team has previous experience with react and it supports Android and IOS app development. It allows for hot reloading which will allow us to rapidly prototype our project without wasting time re-compiling an entire project.

SQLite & PostgreSQL

At this time, we plan to have data specific to users stored locally using SQLite3. A user's planned meals and inventory fall under this category. Additionally, we plan to store recipes locally for quick queries and processing; however, if our recipe database is too big to store locally, we will migrate it to the cloud using PostgreSQL and allow users to query for recipes when planning meals. Our shared fridge feature will utilize Postgres tables to store information about what users are sharing what items in their fridge, and facilitate the shared quantity tracking.

TypeORM

We will use TypeORM for database interaction to connect our application interface with PostgreSQL and SQLite3 and manage function implementation for recipe retrieval and inventory management. TypeORM supports multiple database including both SQLite and PostgreSQL while offering type safety.

5 Engineering Standards

5.1 Project Management

Project feature tracking

Our team is using github projects to track the development of Pocket Pantry in an organized manner. Github projects will allow us to create detailed tasks, assign team members to them, and set deadlines for each task. To facilitate collaboration, our team will and has been conducting weekly meetings to discuss progress, identify any roadblocks, and adjust timelines.

Coding standards

We will follow a consistent coding style across the entire codebase, utilizing tools such as Prettier and ESLint for JavaScript/TypeScript code formatting and linting. We will follow camel case naming conventions for functions and variables.

In React Native, we will use a component based architecture. Our components will be modular and reusable. Each component will encapsulate its logic and UI and we will organize components by functionality into directories.

5.2 Team Roles

Nick Cunningham
Background: App Development Internship and Exposure to Project Management
Project Role: Project Manager and Full-Stack Developer

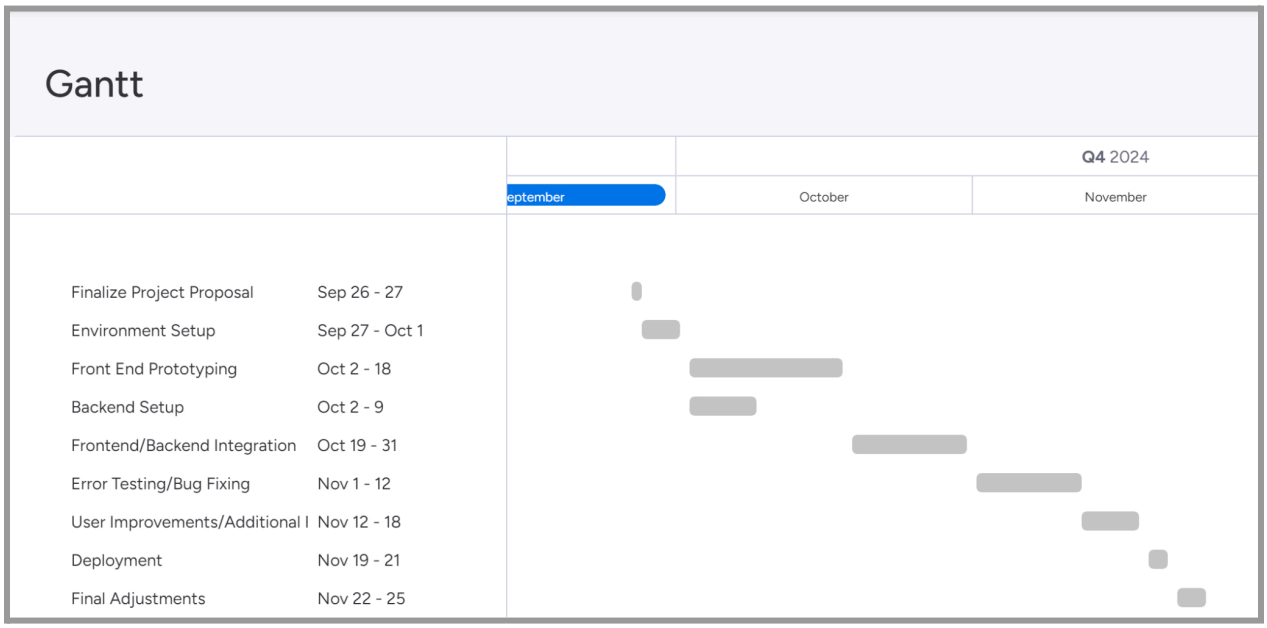
Brian Glasheen
Background: Generally chill dude
Project Role: Full-Stack Developer

Connor Hoffman
Background: Full Stack App Development Experience in Summer Internship
Project Role: Full-Stack Developer

Jacob Miller
Background: Software Development Internship
Project Role: Back-End Developer

Zack Shawver
Background: Computer Science Major and Experience with Design Work
Project Role: Graphics Designer and Front-End Developer

5.3 Schedule of Tasks



The above gantt chart provides a basic overview of our expected timeline and task goals for the duration of this project. We allotted extra time for frontend design, so our team can provide an intuitive design to users without needing instruction. There are a few support features mentioned throughout our document that we hope to include during the User Improvements/Additional Features task period that we think will vastly improve our product's service such as ios compatibility to reach more customers and enable greater recipe filtering.

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